Tilak Kumar Sah

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Summary

Aspiring **Java Developer / Backend Developer** with a strong foundation in **System Design** and **Low-Level Design (LLD)** – applying OOP principles, design patterns, and scalable architecture practices to build production-style backend systems. Hands-on experience across a distributed URL shortener, a task management platform, and a blockchain-based authentication service, backed by a 6-month industry internship as a Java Developer. Working knowledge of **AWS cloud architecture** (EC2, S3, IAM, DynamoDB, VPC), REST API design, caching, and microservices fundamentals.

Education

Tula's Institute, Uttarakhand Technical University — Dehradun, India
B.Tech in Computer Science Engineering; CGPA: 8.0 — *2023 – 2027*
Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Cloud Computing, Software Engineering

Technical Skills

Languages: Java, Python, C++, Dart
System Design: Low-Level Design (LLD), Design Patterns, REST API Development, Microservices, Caching, Load Balancing, Scalability
Backend: Spring Boot, Firebase
Cloud: AWS (EC2, S3, IAM, VPC, DynamoDB, CloudFormation)
Databases: MySQL, MongoDB, DynamoDB
Tools: Git, GitHub, VS Code, Postman

Projects

Scalable URL Shortener | *Java, Spring Boot, MySQL, Redis* — GitHub

- Designed and built a URL shortener service in Java/Spring Boot, generating unique short codes via a base-62 hashing scheme to minimize collision risk at scale.
- Modeled an optimized relational schema and added a caching layer to keep redirect latency low under concurrent load, with a sharding/load-balancing strategy outlined for high-traffic bursts.

Scalable Task Management System | *Java, Firebase, AWS (EC2, S3, DynamoDB)* — GitHub

- Built a modular task management platform with a clean separation between authentication, task, and notification services to support independent scaling.
- Implemented secure Firebase/AWS authentication and architected the backend on AWS (EC2, S3, DynamoDB), adding scheduling and notification features for task reminders.

Blockchain-Based Authentication System | *Java, Blockchain, Cryptography, REST APIs* — GitHub

- Designed a decentralized authentication mechanism that records identity-verification events on a lightweight blockchain ledger, removing reliance on a single trusted authority.
- Applied cryptographic hashing and digital-signature techniques for tamper-resistance, and exposed the verification flow through a REST API layer for external use.

Experience

Java Developer Intern — March – August 2026
Youngminds Technology Solutions Private Limited — *Remote*

- Completed a 6-month internship (AICTE-affiliated program) building Java applications, moving from guided tasks to independently owned feature work over the course of the program.
- Applied core Java and OOP concepts to real project tasks, reinforcing DSA fundamentals alongside hands-on development.
- Collaborated with mentors and peers in a structured program, receiving feedback and iterating on code quality.

Certifications

AWS Certified Cloud Practitioner (2024) – AWS fundamentals: EC2, S3, IAM, DynamoDB.

Microsoft Certified: Azure AI Engineer Associate – AI-102 (2025) – Azure AI, NLP, Computer Vision.

Solved 200+ problems on LeetCode – consistent DSA & problem-solving practice.